

Alicia M. Chun

5312 S. Greenwood Ave., Chicago, IL 60615 | (808) 333-1048 | aliciamchun@uchicago.edu

EDUCATION

The University of Chicago

Chicago, IL

B.S. Major in Computer Science, w/ Specialization in Computer Systems; Minor in Physics

Expected June 2025

Cumulative GPA: 3.73/4.0; President's Scholar, University Scholar Award

Relevant Coursework: Engineering Printed Circuit Boards; Mobile Computing; Computer Networks; Algorithms; Computer Architecture; Electronics

WORK EXPERIENCE

NASA L'SPACE Program, Mission Concept Academy

Remote

Participant - Team Role: Project Manager; 20 hrs/week

Sept. 2024 – Jan. 2025

- Virtual NASA workforce preparation academy that teaches space mission concept formulation.
- Wrote a **183-page Preliminary Design Review** proposal for the NASA mission life cycle of a robotic space mission.
- Managed a team of 15 and organized the team's operations and procedures, such as: scheduling weekly meetings, leading task distribution and progress checks, maintaining consistency across deliverables.

The University of Chicago, Department of Physics

Chicago, IL

Learning Assistant; 10 hrs/week

Sept. 2023 – Now

- Facilitate small labs with ~20 students for the introductory physics and engineering courses.
- Specific courses: Mechanics (Autumn 2023); Electricity and Magnetism (Winter 2024); Waves, Optics, and Heat (Spring 2024); Creative Machines and Innovative Instrumentation (Winter 2025)

Human Computer Integration Lab, Computer Science Department, UChicago

Chicago, IL

Research Associate; 20 hrs/week

Jan. 2024 – Sept. 2024

- Designed and 3D printed wearable devices to use with electric muscle stimulation.
- Programmed microcontrollers (Seeeduino, Arduino Nano, ATmega64) in **C/C++** to control hardware for research.
- Designed and assembled **printed circuit boards** (PCBs) designed to generate electrical impulses used for muscle stimulation using **KiCad**.

Institute for Astronomy, University of Hawai'i at Mānoa

Honolulu, HI

Astrophysics REU Fellow; 40 hrs/week

Jan. 2024 – Sept. 2025

- Conducted statistical analyses of 10,000+ characteristics of M-giant stars using data from TESS and Kepler in **Python3**.
- Cross-compared oscillation periods obtained from space-based telescopes to ground-based transient surveys.
- Found the frequency of maximum oscillation power of 10,000+ M-giants using ASAS-SN survey.
- Talk: Chun, A., Saunders, N., Huber, D. (2023) Testing the Asteroseismic Detection Limits of Ground-Based Transient Survey using Kepler and TESS¹

Subaru Telescope, National Astronomical Observatory of Japan

Hilo, HI

Research Intern; 40 hrs/week

Jul. 2022 – Sept. 2022

- Characterized the AO3000, a new continuous surface deformable mirror that had 4,096 actuators.
- Created an automated data collection system controlling 1) movement of the deformable mirror, 2) data collection of laser interferometer, 3) data processing and saving in **Python3**, and 4) presented to parent company, ALPAO.
- Publication: Lozi, L., Ahn, K., et al. including Chun, A., (2024). AO3k at Subaru: First on-sky results of the facility extreme-AO. arXiv:2407.19188

UNIVERSITY PROJECTS

Mobile Theremin

Mar. 2024

- Demo: Awarded 2nd place project in class²
- Created a mobile theremin Apple application by using the True Depth sensor to control volume and the gyroscope rotation on the apple watch to control pitch using Swift.

RhythEMS — Wearable 'Loop Pedal'

Mar. 2024

- Worked with a partner, Elena Hertel, to design and build a wearable device that incorporates EMS and a loop pedal.
- Developed the circuitry for the device, created the final protoboard, and wrote the code to control the EMS output.

¹ <https://www.youtube.com/watch?v=hNwKU7kQk0Q>

² <https://youtube.com/shorts/adIGvrU0zSU?si=Q-R5AU7U-Xabkdec>

- Demo: The user records a rhythm with a drumstick, and the device plays this rhythm on a loop using EMS.³

LEADERSHIP EXPERIENCE

Le Vorris and Vox Circus

President, Acrobatics Instructor

Chicago, IL

March 2022 – Present

- Direct, perform, and write three circus shows, each with 20+ performers, every year for three years for 300+ people.
- Reserve spaces, budgeting finances, bridge communication between faculty, performers, and tech staff.

Association for Computing Machinery, UChicago

Event Coordinator

Chicago, IL

Jan. 2023 – Present

- Book weekly rooms and schedule events with the board.

National Space Society, SpacEdge

Metcalf Intern

Chicago, IL

Sept. 2024 – Jan. 2025

- Research and develop courses for topics such as microgravity and planetary defense.

ADDITIONAL

Technical Skills: Git, Arduino IDE, AutoCAD, KiCad, Siemens NX, Sklearn (ML/AI)

Programming Skills: Java, C/C++, Python3, Swift, HTML/CSS

Languages: Fluent in English; Intermediate Proficiency in Spanish; Beginner German, Portuguese

Misc: Tennis, Piano, Saxophone, Aerial Silks, Lyra, Partner Acrobatics

³ <https://www.youtube.com/watch?v=A-OjZe4ul9s>